Jesús Emilio Domínguez Russell

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Education

University of Maryland, College Park — College Park, MD, USA
Ph.D. in Mathematics – Advisor: Amin Gholampour. — August 2021–Present

Universidad Autónoma de Sinaloa — Culiacán, Sinaloa, México
B.Sc. in Mathematics. — August 2016–July 2021

Publications/Preprints

[1] E. Dominguez and A. Gholampour, *Virtual cycles of 3-term complexes and the hilbert schemes of surfaces*, 2026. arXiv: 2606.29542 [math.AG].

[2] O. Arizmendi and E. Dominguez, "Barabasi–albert trees are hypoenergetic", *Boletín de la Sociedad Matemática Mexicana*, vol. 28, no. 3, p. 72, Sep. 2022, ISSN: 2296-4495.

[3] J. E. Domínguez and C. Segovia, "Extending free actions of finite groups on surfaces", *Topology and its Applications*, vol. 305, p. 107 898, 2022, ISSN: 0166-8641.

Teaching

- **Instructor for the University of Maryland** — Summer 2023, Summer 2024
  Calculus I (MATH140)
- **Teaching Assistant for the University of Maryland** — Spring 2021–Fall 2026
  Introduction to Linear Algebra (MATH240), Calculus I (MATH140), Elementary Calculus I (MATH120), Precalculus (MATH115)
- **Grader for the University of Maryland** — Fall 2024, Summer 2026
  Advanced Calculus II (MATH411), Applications of Linear Algebra (MATH401).

Seminar Talks

- **Virtual cycles of 3-term complexes** — October 2026
  Algebraic geometry seminar, at the Ohio State University
- **Virtual cycles of 3-term complexes** — October 2026
  Algebraic geometry seminar, at the University of Maryland

Student Seminar Talks

- **Relative GW theory** — March 2026
  RIT on Algebraic Geometry – Logarithmic Geometry, at University of Maryland
- **Virtual Classes and GW theory** — February 2026
  RIT on Algebraic Geometry – Logarithmic Geometry, at University of Maryland
- **Moduli of vector bundles and Brill-Noether Theory** — February 2025
  RIT on Algebraic Geometry – K3 Surfaces, at University of Maryland
- **Intersections in Algebraic Geometry** — May 2024
  Student Seminar, at Universidad Autónoma de Sinaloa, Mexico
- **The Inverse Galois Problem** — April 2024
  2024 Seymour Goldberg and Monroe Martin Spotlight Talks, at University of Maryland

Mentoring

- **Counselor at Brin Maryland Mathematics Camp** — Summer 2024, Summer 2025, Summer 2026
  Courses in Knot Theory and Mathematical Biology
- **Mentor for Undergraduate Directed Reading Program at UMD** — Spring 2023, Fall 2025

Topics: "Modular forms and their applications", "Space-filling curves"

- **Coach for Sinaloa state team in the Mexican Math Olympiads** 2016–2020

Awards

- **Herbert A. Hauptman Fellow** 2025, 2026
- **Dean's Fellowship, University of Maryland** 2021–2023